

Teacher card

Climate change causes and effects / GROW_S2_W03

Outcome: Investigate causes and effects of climate change.

Evidence to retain: F: evidence-based explanation, targeted questioning and two exit questions.

Forty minute route

Arrival 4 min; starter 3; model 6; shared task 7; check 2; independent 12; review 3; exit 3. Pause time is flexible.

Preparation

Choose one independent route. Print the evidence and shared cards, required visual or data sheet, route worksheet and exit. The organiser is one separate A4 page. Keep answers with staff.

Use evidence to explain problems and feasible responses. Avoid asking pupils to carry responsibility for solving global climate change.

Access and regulation

Chunk sources and tasks; pre-teach vocabulary with images. Offer sentence stems, organisers, oral rehearsal and typing. Start with one modelled example; extend through a second source, comparison or supported judgement. Check the learner can begin within two minutes, then fade prompts.

Offer a predictable start, private response and agreed pause. No cold calling or forced personal disclosure. Rejoin at one clearly named next step. Keep teen-appropriate examples and the same learning goal across response modes.

Use sensitive content warnings where relevant. Do not ask for family details or personal experiences.

Assessment

Use questioning and the exit response to decide the next step. Record the learner's individual contribution and the access support used.

UAS: no approved unit code is supplied. Match the learner's evidence to the centre-selected unit and exact criteria before submission. This pack makes no automatic award claim.

Model and answers

Read these pages before teaching. Hide the model during independent assessment.

Question: What happens when fossil fuels are burned?

Evidence: B Human influence: Burning fossil fuels adds carbon dioxide. Greenhouse gases slow heat loss to space. Human activity is the main cause of recent global warming.

Think aloud: Carbon dioxide is added to the atmosphere, strengthening the warming influence.

Check: How do emissions reduction and shade differ? The first tackles a cause; shade can reduce exposure to heat.

Arrival answers

1. A tsunami.
2. Weather patterns over a long period
3. Carbon dioxide is added to the atmosphere, strengthening the warming influence.
4. The first tackles a cause; shade can reduce exposure to heat.

Shared activity reasoning

Exit responses

What happens when fossil fuels are burned? Carbon dioxide is added to the atmosphere, strengthening the warming influence.

How do emissions reduction and shade differ? The first tackles a cause; shade can reduce exposure to heat.

Teaching and evidence record

Supplied scheme of work teaching sequence: Use climate maps and a short data set to spot a pattern. Sort cause/effect cards together; independently explain a supported link and compare one response by people or places.

Look for

F: evidence-based explanation, targeted questioning and two exit questions.

Accept accurate equivalents, including speech or exact scribing. Check the concept and use of evidence rather than length or handwriting. Record access support separately from content prompting.

BUILD: a concrete example with a reason or demonstrated action. GROW: explain or compare with a relevant detail. LAUNCH: connect evidence and reasoning; use a limitation when the outcome requires it. Keep RE responses non-GCSE.

Media

Original captioned teaching model made for this pack. It demonstrates the supplied reasoning; it is not historical footage, survivor testimony or a documentary. The full transcript and factual source credits are included.

Use manual steps for a calm pace. The silent MP4 has visible captions and a matching text transcript. External archive media is optional and should be previewed before class.

Source credits

NASA Causes of climate change <https://science.nasa.gov/climate-change/causes/>

Source access was checked on 19 September 2026. Restricted links are listed in Sources_and_checks.html; a restriction here is not proof that the page is dead. Offline teaching material remains available.